INFOSYS SE QUANT VAULT • TIME, SPEED, DISTANCE & WORK

Relative Speed, Trains, Boats, Pipes & Cisterns, Work Equivalence • Total Solved Questions: 45

[Campus Monk Day 01 Mock Test]
Topic: *Quantitative Aptitude*

Q1. A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ to finish the work exactly on time?

(A) 40

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 01 Mock Test]
Topic: *Quantitative Aptitude*

Q2. Note : Ans in days

(A) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 01 Mock Test]
Topic: *Quantitative Aptitude*

Q3. In a river, the ratio of the speed of stream and speed of a boat A in still water is 2 : 5. Again, in the same river ratio of the speed of stream and speed of another boat B in still water is 3 : 4. What is the ratio of the speeds of the first boat to the second boat in still water ?

(A) 10 : 7
(B) 15 : 8
(C) 4 : 3
(D) 5 : 4

Correct Answer: Option (B) : 15 : 8

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : 15 : 8.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q4. Two ants of length 1 cm and 1.2 cm crawl in opposite directions with average speeds of 2 and 3 mm per second respectively. How many seconds will they take to cross each other?

(A) 0.44 sec
(B) 44 sec
(C) 6.4 sec
(D) ---

Correct Answer: Option (D) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : ---.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q5. Two trains are on the same line, 100 miles apart, heading towards each other, each traveling at 20 miles/h. A bee starts flying between the trains when the distance between two trains is 100 km. The bee first flies from first train to second train. Once it reaches the second train, it immediately flies back to the first train ... and so on until trains collide. Calculate the total distance travelled by the bee, if Speed of bee is 40 miles/h

(A) ---
(B) 150
(C) 170
(D) 200

Correct Answer: Option (A) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : ---.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q6. Quantity 1: Speed of boat in still water, if a man can travel 54 km downstream in 9 hours and 40 km upstream in 10 hours. Quantity II: Speed of boat in still water, if a man can travel 45 km downstream in 9 hours and the speed of stream is 1 kmph.

(A) - (B)
(C) ---
(D) - (E)

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q7. A person can complete a job in 120 days. He works alone on Day 1. On Day 2, he is joined by another person who also can complete the job in exactly 120 days. On Day 3, they are joined by another person of equal efficiency. Like this, everyday a new person with the same efficiency joins the work. How many days are required to complete the job?

(A) 20
(B) ---
(C) 25
(D) NOT

Correct Answer: Option (B) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : ---.

[Campus Monk Day 02 Mock Test]
Topic: *Quantitative Aptitude*

Q8. The Ghaziabad-Hapur-Meerut EMU and the Meerut-Hapur-Ghaziabad EMU start at the same time from Ghaziabad and Meerut and proceed towards each other at 16 km/hr and 21 km/hr, respectively. When they meet, it is found that one train has travelled 60 km more than the other. The distance between two stations is:

(A) 445 km
(B) ---
(C) 440 km
(D) 450 km

Correct Answer: Option (B) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : ---.

[Campus Monk Day 03 Mock Test]
Topic: *Quantitative Aptitude*

Q9. "A" can complete a piece of work in 8 days,while "B" is 20% more efficient than him. A works for 2 days,then B works for 1 day, and the cycle continues like this.in how many days would the work be completed?

(A) ---
(B) 7 days
(C) 7(7/9)
(D) not

Correct Answer: Option (A) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : ---.

[Campus Monk Day 06 Mock Test]
Topic: *Quantitative Aptitude*

Q10. A boy walked along two adjacent sides of a rectangular field. If he had walked along the diagonal, then he would have saved a distance equal to one-fourth of the larger side. The ratio of the smaller to the larger side is:

(A) 11:18
(B) 18:11
(C) 7:24
(D) 5:12

Correct Answer: Option (C) : 7:24

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : 7:24.

[Campus Monk Day 07 Mock Test]

Topic: *Quantitative Aptitude*

Q11. There are two pipes in a tank. Pipe A is for filling the tank and pipe b is for emptying the tank. If A can fill the tank in 10 hours and b can empty the tank in 15 hours then find how many hours will it take to completely fill a half empty tank?

(A) A. 30 hours
(B) A. 15 hours
(C) A. 20 hours
(D) A. 33.33 hours

Correct Answer: Option (B) : A. 15 hours

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : A. 15 hours.

[Campus Monk Day 07 Mock Test]

Topic: *Quantitative Aptitude*

Q12. The total time taken by a boat to cover 'D+25' km downstream and same distance in upstream is 20 hours. If speed of boat is 15 km/hr. in still water and speed of stream is 40% less than speed of boat, then find the value of D

(A) 66 km
(B) 78 km
(C) 62km
(D) 71 km

Correct Answer: Option (D) : 71 km

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : 71 km.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q13. Question : If Raman drives his bike at a speed of 24 km/h, he reaches his office 5 minutes late. If he drives the bike at a speed of 30 km/h, he reaches his office 4 minutes early. How much time (in minutes) will he take to reach his office at a speed of 27 km/h?

(A) 40
(B) 55
(C) 50
(D) 45

Correct Answer: Option (A) : 40

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : 40.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q14. Question : Avra travels from A to B at 74 km/h and returns from B to A through the same route at 111 km/h. What was his average speed during the two-way journey?

(A) 78.88 km/hr
(B) 88.88 km/hr
(C) 98.88 km/hr
(D) 108.88 km/hr

Correct Answer: Option (B) : 88.88 km/hr

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : 88.88 km/hr.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q15. Question : A can do 1/4 of a work in 3 days and B can do 1/6 of the same work in 4 days, how much will A get if both work together and are paid Rs. 180 in all?

(A) 150 Rs
(B) 100
(C) 120
(D) 160

Correct Answer: Option (C) : 120

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : 120.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q16. Question : Two boys Rishi and Vamsi start at the same time to ride from Lucknow to Kanpur that is 95 km away. Rishi travels 5 km/h slower than Vamsi. Vamsi reaches Kanpur and immediately start to travel back. On his return journey he meets Rishi who is 25 km away from Kanpur. Find Rishi's speed(Km/hr)

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q17. Question : The two trains are coming towards each other and cross a signal pole in 30 seconds and 20 seconds respectively and they cross each other in 24seconds. Find the ratio of the speed of these two trains?

(A) 2:3
(B) 3:5
(C) 4:5
(D) 3:4

Correct Answer: Option (A) : 2:3

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : 2:3.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q18. P started from his house towards west. After walking a distance of 25 m. He turned to the right and walked 10 m. He then again turned to the right and walked 15 m. After this he is to turn right at 135 degree and to cover 30 m. In which direction P is going?

(A) West
(B) South
(C) South-East
(D) South-West

Correct Answer: Option (D) : South-West

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : South-West.

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q19. Read the passage and answer the questions that follow : Once upon a time, in a quaint little village nestled in a picturesque valley, there lived a young girl named Lily. She was known for her adventurous spirit and her deep curiosity. One sunny afternoon, while exploring the woods that bordered the village, Lily stumbled upon a hidden clearing. In the center of this secret meadow, she found an old oak tree with a peculiar nook. In that nook, half-buried in the earth, was a mysterious locket, ornate and gleaming with intricate designs. Lily carefully unearthed it, and as she held it in her hand, she felt a strange connection to this forgotten piece of jewelry. Back in her cottage, Lily examined the locket more closely. It was made of silver with a hint of gold, and its surface was adorned with delicate engravings of leaves and flowers. On the front, there was an image of a crescent moon, and on the back, an engraving read, "To the one who seeks the truth." The locket was sealed tightly, and Lily couldn't open it no matter how hard she tried. Determined to solve the mystery, Lily embarked on a quest to find someone who could open the locket. She asked the village blacksmith, the wise old woman who lived on the hill, and even the town's mayor, but none of them could unlock its secrets. Feeling a bit discouraged, Lily spent her days reading about ancient relics and searching for clues in the library. One day, while immersed in an old book about mystical lockets, Lily came across a passage that mentioned the existence of a "key of the heart." It said that this key could open the most stubborn of locks, but it was hidden within each person, waiting to be discovered. This newfound knowledge gave Lily a renewed sense of hope and determination. The next morning, Lily decided to visit the village's recluse, an elderly woman rumored to have deep knowledge of the heart. The recluse lived in a remote cottage surrounded by a lush garden. Lily knocked on the door and explained her quest. The old woman listened attentively and finally agreed to help. She told Lily that the key to opening the locket was hidden within her heart and that she needed to meditate on her truest desires and intentions. Lily followed the recluse's advice, sitting beneath the ancient oak tree where she had found the locket. She closed her eyes, focused on her heart's deepest desires, and as she did, she felt the locket warming in her hand. When she opened her eyes, she found the locket had sprung open, revealing a tiny, folded parchment inside. With trembling hands, she carefully unfolded it and read the message: "The truth lies in the journey, not the destination." The mystery of the locket was solved, but Lily realized that the real treasure was the journey of self-discovery she had undertaken. She understood that her quest had led her to understand her heart's desires and the power of her own determination. Question 1- What did Lily discover in the hidden clearing in the woods?

(A) A hidden garden
(B) An old oak tree
(C) A mysterious book
(D) A magic wand

Correct Answer: Option (B) : An old oak tree

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (B) : An old oak tree.

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q20. Read the passage and answer the questions that follow : Once upon a time, in a quaint little village nestled in a picturesque valley, there lived a young girl named Lily. She was known for her adventurous spirit and her deep curiosity. One sunny afternoon, while exploring the woods that bordered the village, Lily stumbled upon a hidden clearing. In the center of this secret meadow, she found an old oak tree with a peculiar nook. In that nook, half-buried in the earth, was a mysterious locket, ornate and gleaming with intricate designs. Lily carefully unearthed it, and as she held it in her hand, she felt a strange connection to this forgotten piece of jewelry. Back in her cottage, Lily examined the locket more closely. It was made of silver with a hint of gold, and its surface was adorned with delicate engravings of leaves and flowers. On the front, there was an image of a crescent moon, and on the back, an engraving read, "To the one who seeks the truth." The locket was sealed tightly, and Lily couldn't open it no matter how hard she tried. Determined to solve the mystery, Lily embarked on a quest to find someone who could open the locket. She asked the village blacksmith, the wise old woman who lived on the hill, and even the town's mayor, but none of them could unlock its secrets. Feeling a bit discouraged, Lily spent her days reading about ancient relics and searching for clues in the library. One day, while immersed in an old book about mystical lockets, Lily came across a passage that mentioned the existence of a "key of the heart." It said that this key could open the most stubborn of locks, but it was hidden within each person, waiting to be discovered. This newfound knowledge gave Lily a renewed sense of hope and determination. The next morning, Lily decided to visit the village's recluse, an elderly woman rumored to have deep knowledge of the heart. The recluse lived in a remote cottage surrounded by a lush garden. Lily knocked on the door and explained her quest. The old woman listened attentively and finally agreed to help. She told Lily that the key to opening the locket was hidden within her heart and that she needed to meditate on her truest desires and intentions. Lily followed the recluse's advice, sitting beneath the ancient oak tree where she had found the locket. She closed her eyes, focused on her heart's deepest desires, and as she did, she felt the locket warming in her hand. When she opened her eyes, she found the locket had sprung open, revealing a tiny, folded parchment inside. With trembling hands, she carefully unfolded it and read the message: "The truth lies in the journey, not the destination." The mystery of the locket was solved, but Lily realized that the real treasure was the journey of self-discovery she had undertaken. She understood that her quest had led her to understand her heart's desires and the power of her own determination. Question 2- What was inscribed on the back of the mysterious locket?

(A) "To the one who seeks adventure"
(B) "To the one who seeks answers"
(C) "To the one who seeks the truth"
(D) "To the one who seeks gold"

Correct Answer: Option (C) : "To the one who seeks the truth"

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : "To the one who seeks the truth".

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q21. Read the passage and answer the questions that follow : Once upon a time, in a quaint little village nestled in a picturesque valley, there lived a young girl named Lily. She was known for her adventurous spirit and her deep curiosity. One sunny afternoon, while exploring the woods that bordered the village, Lily stumbled upon a hidden clearing. In the center of this secret meadow, she found an old oak tree with a peculiar nook. In that nook, half-buried in the earth, was a mysterious locket, ornate and gleaming with intricate designs. Lily carefully unearthed it, and as she held it in her hand, she felt a strange connection to this forgotten piece of jewelry. Back in her cottage, Lily examined the locket more closely. It was made of silver with a hint of gold, and its surface was adorned with delicate engravings of leaves and flowers. On the front, there was an image of a crescent moon, and on the back, an engraving read, "To the one who seeks the truth." The locket was sealed tightly, and Lily couldn't open it no matter how hard she tried. Determined to solve the mystery, Lily embarked on a quest to find someone who could open the locket. She asked the village blacksmith, the wise old woman who lived on the hill, and even the town's mayor, but none of them could unlock its secrets. Feeling a bit discouraged, Lily spent her days reading about ancient relics and searching for clues in the library. One day, while immersed in an old book about mystical lockets, Lily came across a passage that mentioned the existence of a "key of the heart." It said that this key could open the most stubborn of locks, but it was hidden within each person, waiting to be discovered. This newfound knowledge gave Lily a renewed sense of hope and determination. The next morning, Lily decided to visit the village's recluse, an elderly woman rumored to have deep knowledge of the heart. The recluse lived in a remote cottage surrounded by a lush garden. Lily knocked on the door and explained her quest. The old woman listened attentively and finally agreed to help. She told Lily that the key to opening the locket was hidden within her heart and that she needed to meditate on her truest desires and intentions. Lily followed the recluse's advice, sitting beneath the ancient oak tree where she had found the locket. She closed her eyes, focused on her heart's deepest desires, and as she did, she felt the locket warming in her hand. When she opened her eyes, she found the locket had sprung open, revealing a tiny, folded parchment inside. With trembling hands, she carefully unfolded it and read the message: "The truth lies in the journey, not the destination." The mystery of the locket was solved, but Lily realized that the real treasure was the journey of self-discovery she had undertaken. She understood that her quest had led her to understand her heart's desires and the power of her own determination. Question 3- Who did Lily seek help from in opening the locket?

(A) The village blacksmith
(B) The wise old woman on the hill
(C) The town's mayor
(D) All of the above

Correct Answer: Option (D) : All of the above

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : All of the above.

[Campus Monk Day 08 Mock Test]

Topic: *Quantitative Aptitude*

Q22. Question : The committee urged all departments to work in a _____ manner to achieve the organization's long-term goals.

(A) coordinated
(B) casual
(C) negligent
(D) contradictory

Correct Answer: Option (A) : coordinated

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : coordinated.

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q23. A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ to finish the work exactly on time?

(A) 40

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q24. Note : Ans in days

(A) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is See Logic.

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q25. In a certain code, '786' means 'study very hard', '958' means 'hard work pays' and '645' means 'study and work'. Which of the following is the code 'very'?

(A) 8
(B) 6
(C) 7
(D) Cannot be determined

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : 7.

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q26. They all laughed as they realized that although they had gone off in different directions from working at home to managing companies they were experiencing _____ feeling

(A) Jubilant
(B) Remorseful
(C) Similar
(D) Strange

Correct Answer: Option (A) : Jubilant

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (A) : Jubilant.

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q27. To my horror, (a)/ my wedding dress looked (b)/ strain, torn (c)/ and wrinkled (d)/

(A) Part a
(B) Part b
(C) Part c
(D) Part d

Correct Answer: Option (C) : Part c

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : Part c.

[Campus Monk Day 09 Mock Test]

Topic: *Quantitative Aptitude*

Q28. He was very tired as he is working since 6 O' clock in the morning.

(A) he was working
(B) he had been working
(C) > ■
(D) he will be working

Correct Answer: Option (C) : > ■

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : > ■.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q29. Direction : In these question you have two brief passages with six questions following the passage. Read the passage carefully and choose the best answer to each question out of the four alternatives. Passage – 1 Education, as Mahatma Gandhi described it, "is the tool for the development of consciousness and reconstitution of society". Since Independence, India has stressed reforming and restructuring the educational system as part of state intervention. The National Policy on Education (NPE), 1986, which is hailed as a landmark in the Indian educational system, provided a comprehensive framework to guide the development of education in the country. The NPE and its Programme of Action was again updated in 1992 through similar consensual process involving all the State governments, resource organisations and educationists. What has been worrying the critics and educationists alike is the non- fulfilment of one of the NPE objectives: education of girl. It has been stated in the NPE that the 'Education For All' meet should have a strong gender focus as Education For All by definition is gender inclusive. . The National Policy on Education provided

(A) guidelines for the comprehension of education in the country

(B) guidance material to develop education in the country

(C) ---

(D) comprehensive development of education in the country

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q30. Direction In these question you have two brief passages with six questions following the passage. Read the passage carefully and choose the best answer to each question out of the four alternatives. Passage – 1 Education, as Mahatma Gandhi described it, "is the tool for the development of consciousness and reconstitution of society". Since Independence, India has stressed reforming and restructuring the educational system as part of state intervention. The National Policy on Education (NPE), 1986, which is hailed as a landmark in the Indian educational system, provided a comprehensive framework to guide the development of education in the country. The NPE and its Programme of Action was again updated in 1992 through similar consensual process involving all the State governments, resource organisations and educationists. What has been worrying the critics and educationists alike is the non- fulfilment of one of the NPE objectives: education of girl. It has been stated in the NPE that the 'Education For All' meet should have a strong gender focus as Education For All by definition is gender inclusive. According to the passage, critics and educationists are worried that

(A) the education of girls is one of the objectives of NPE.

(B) the objectives of NPE have not been fulfilled.

(C) non-fulfilment of NPE leads to the education of girls.

(D) ---

Correct Answer: Option (D) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : ---.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q31. Direction In these question you have two brief passages with six questions following the passage. Read the passage carefully and choose the best answer to each question out of the four alternatives. Passage – 1 Education, as Mahatma Gandhi described it, "is the tool for the development of consciousness and reconstitution of society". Since Independence, India has stressed reforming and restructuring the educational system as part of state intervention. The National Policy on Education (NPE), 1986, which is hailed as a landmark in the Indian educational system, provided a comprehensive framework to guide the development of education in the country. The NPE and its Programme of Action was again updated in 1992 through similar consensual process involving all the State governments, resource organisations and educationists. What has been worrying the critics and educationists alike is the non- fulfilment of one of the NPE objectives: education of girl. It has been stated in the NPE that the 'Education For All' meet should have a strong gender focus as Education For All by definition is gender inclusive. According to the passage, 'a strong gender focus' means

(A) a focus on the strength of gender

(B) a focus on strong gender

(C) ---

(D) a focus on male-female ratio

Correct Answer: Option (C) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (C) : ---.

[Campus Monk Day 09 Mock Test]

Topic: Quantitative Aptitude

Q32. Direction In these question you have two brief passages with six questions following the passage. Read the passage carefully and choose the best answer to each question out of the four alternatives. Passage – 1 Education, as Mahatma Gandhi described it, "is the tool for the development of consciousness and reconstitution of society". Since Independence, India has stressed reforming and restructuring the educational system as part of state intervention. The National Policy on Education (NPE), 1986, which is hailed as a landmark in the Indian educational system, provided a comprehensive framework to guide the development of education in the country. The NPE and its Programme of Action was again updated in 1992 through similar consensual process involving all the State governments, resource organisations and educationists. What has been worrying the critics and educationists alike is the non- fulfilment of one of the NPE objectives: education of girl. It has been stated in the NPE that the 'Education For All' meet should have a strong gender focus as Education For All by definition is gender inclusive. . In the passage, the author makes a plea for

(A) free education

(B) universal education

(C) the education of men

(D) ---

Correct Answer: Option (D) : ---

■ **Step-by-Step Logic:** Solving using quantitative formula derivation confirms the answer is Option (D) : ---.

[College ITM Mock Paper 1]

Topic: Mathematical Critical Thinking

Q33. A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, … (one person per day, starting with A). On which day is the job completed?

(A) 16th

(B) 17th

(C) 18th

(D) 19th

Correct Answer: Option (A) : 16th

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 1]
Topic: Mathematical Critical Thinking
Q34. Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:
(A) 90 km/h
(B) 108 km/h
(C) 120 km/h
(D) 72 km/h
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]
Topic: Mathematical Critical Thinking
Q35. A and B together can complete a task in 15 days. They worked together for 5 days, after which A left and B completed the remaining work alone in 20 more days. In how many days can A alone complete the task?
(A) 20 days
(B) 24 days
(C) 25 days
(D) 30 days
Correct Answer: Option (A) : 20 days
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 2]
Topic: Mathematical Critical Thinking
Q36. A boat's speed in still water is 15 km/h. It travels 60 km downstream and returns upstream to the starting point, taking a total of 9 hours. Find the speed of the stream.
(A) 4 km/h
(B) 5 km/h
(C) 6 km/h
(D) 3 km/h
Correct Answer: Option (A) : 4 km/h
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q37. A dealer's cost price for a mixture of two rice varieties (mixed 3:2 by weight at ■40/kg and ■60/kg) works out to ■48/kg. He marks up the price by 25% and sells it, but uses a false weight that gives only 900 g for every 1 kg paid for. Find his actual profit percentage.
(A) 38.9%
(B) 35%
(C) 41.7%
(D) 33.3%
Correct Answer: Option (A) : 38.9%
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q38. Pipe A can fill a tank in 12 hours, Pipe B in 15 hours, and a leak can empty the full tank in 20 hours. All three are opened together for 4 hours, after which the leak is plugged. How much more time do A and B together need to fill the remaining tank?
(A) 4 hours
(B) 3 hours
(C) 5 hours
(D) 6 hours
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 3]
Topic: Mathematical Critical Thinking
Q39. In a 1000 m race, A beats B by 100 m, and in a separate 1000 m race, B beats C by 100 m (each runner maintains a constant speed across races). By how many metres does A beat C in a 1000 m race between them?
(A) 190 m
(B) 200 m
(C) 180 m
(D) 210 m
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q40. A, B and C can complete a work alone in 20, 30 and 60 days respectively. All three start together. After 4 days, B leaves; after 3 more days, C also leaves, and A finishes the rest alone. In how many total days is the work completed?
(A) 15 days
(B) 14 days
(C) 16 days
(D) 13 days
Correct Answer: Option (A) : 15 days
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 4]
Topic: Mathematical Critical Thinking
Q41. Two trains, 300 km apart, move towards each other on the same track. Train A departs at 8:00 AM at 60 km/h; Train B departs at 8:30 AM at 90 km/h. At what time do they meet?
(A) 10:18 AM
(B) 10:12 AM
(C) 10:06 AM
(D) 10:24 AM
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q42. A, B and C can individually complete a work in 15, 20 and 30 days. They work together, and the total payment of ■9,000 is distributed in proportion to each person's work contribution. How much should C be paid?
(A) ■2,000
(B) ■1,800
(C) ■2,200
(D) ■2,500
Correct Answer: Option (A) : ■2,000
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 5]
Topic: Mathematical Critical Thinking
Q43. Two runners start from the same point on a 600 m circular track at the same time, in opposite directions, at 8 m/s and 10 m/s. After how many seconds will they both simultaneously be back at the starting point for the first time?
(A) 300 s
(B) 200 s
(C) 150 s
(D) 600 s
Correct Answer: See Logic
■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: *Mathematical Critical Thinking*

Q44. A is twice as efficient as B, and B is three times as efficient as C. Working together, all three complete a job in 9 days. In how many days can C alone complete it?

(A) 90 days
(B) 81 days
(C) 60 days
(D) 72 days

Correct Answer: Option (A) : 90 days

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.

[College ITM Mock Paper 6]

Topic: *Mathematical Critical Thinking*

Q45. A 180 m long train passes a man walking at 6 km/h in the same direction in 12 seconds. How long will this train take to completely cross a 420 m long platform?

(A) 36 s
(B) 30 s
(C) 40 s
(D) 42 s

Correct Answer: Option (A) : 36 s

■ **Step-by-Step Logic:** Verified by ITM Mock Answer Key.